

Can you *prove* that?

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The world was created *Last Thursday*. That's right. Earth, the planets of our solar system, the Sun. All of it came into fruition last Thursday. That's not even considering the rest of the universe which is a wide expanse of stars, planets and galaxies that stretch over 93 billion light-years into the distance!

On top of all of this, there is everything that has been created by humankind. Centuries of work, creation, ideas, and constructs. Society as we know it. Trillions of pages of writing, millions of hours of video footage and a ludicrous amount of photographs.

There are also the memories. We all carry a backlog of memories from our lives. The good, the happy and the joyous. But also the painful, the sad, and those other ones that keep you up at 3am and make you cringe. At least if the universe only started last week, that eradicates all those cringey embarrassing moments.

Now I know that you'll have some queries & questions. *But what about New Year's Eve 2024, I remember it so vividly?* Nope. Sorry, that didn't happen. *Hang on, but I bought my phone last year and that's how I'm reading this now?* How can you prove that you bought it last year? All you have is the memory, right? *No. I've got the receipt.* Okay, but that could have been fabricated or created along with everything else. Take a moment and see if you can think of any other things that might contradict this theory. Actually do it, think!

I bet some of you have just thought, *But what about any people that were alive before last Thursday, like Einstein. Are you saying he could be a completely fabricated memory?* Exactly, that's the thing with this theory. We only have photos and papers and memories of Einstein. We know for a fact that those first two can be fabricated with AI and a bit of time, but that latter one is harder. Memories can't be adapted and altered, yet. Unless everything was created last Thursday, and designed to look as though the universe had been around for 13.8 billion years.

At this point, I am sure many of you are quite thrown and I do sincerely apologise if I have started any existential panics or concerns. But this theory is quite an interesting one, and leads perfectly on to the topic I want to delve into today. *Last Thursdayism* is a theory that originated in the late 19th century when contradictions appeared in the story of Creation. While the Bible claims that Earth has been around for around 6 to 10 thousand years, scientists argue that it is significantly older. A whopping 4.53 billion years in fact! Philosophers of the day then said "*if you claim that Earth is 6000 years old and yet looks billions of years old, who is to say it wasn't made last Thursday?*"

The crux of this theory, is that you are unable to prove it to be false. For every example you give me, of life before last Thursday, I can find an example of how it would fit within the theory:

- *Photographs*. They could have been created to give an impression of life before a week ago. Think, Black Mirror, One Hour Photo or Gone Girl.
- *Notes/books/writings*. These are even easier to forge or falsify than photos. A supposed Creator would definitely be able to make these documents to populate our world. It's not too dissimilar to what Lee Israel did, in trying to craft a fake world of letters between dead celebrities.
- *Fossils & Artefacts*. Just research the 'Piltdown Fly'.
- *Memories*. Severance does a brilliant job at illustrating how memories can be adapted, as does Before I Go To Sleep. Although it may not be the easiest, it is doable. Showmen, such as Derren Brown, are able to adapt your memories, so who's to say a being with enough power to design a universe as intricate as ours, wouldn't be able to.

How does this relate to Mathematics?

The idea of absolute proof, is something that is unique to the pursuit of mathematics. Where lawyers & the judicial systems need a burden of proof that goes beyond reasonable doubt, before convicting a criminal, and scientists carry out a series of tests and experiments to gather a multitude of data, before deciding on a hypothesis - mathematicians need to have complete certainty when proving any statement.

The Last Thursday theory cannot be proven or disproven, mathematically, as there is not complete certainty. I can solve any problem that you present to me, but I cannot definitively show that the theory is accurate for *every* problem. Nor is there any evidence to prove that the theorem is completely false. In other words, I cannot find a *contradiction*.

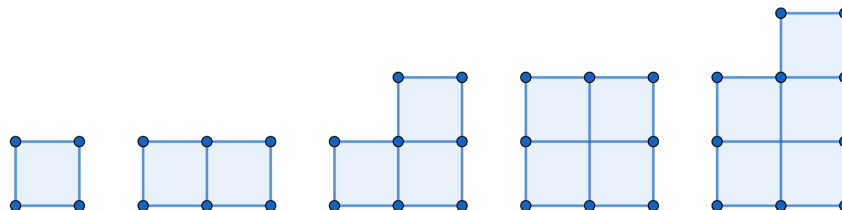
Let's come up with a hypothesis to either prove or disprove to illustrate this idea:

Theorem: The sum of two *odd* numbers is always an *even* number.

At this point, we could start by testing our hypothesis and adding some numbers together. $5 + 7 = 12$ or $13 + 3 = 16$. Both of these work, so it looks like my statement might be true. Have I proven it?

Clearly, there are many many more odd numbers that I haven't tried. Does it still work with 17? Or what about 53? Or, one of my favourites, 111? The difficulty with doing it manually and testing each and every number, is that there are an infinite amount of numbers. Further, there are an infinite amount of numbers to pair this infinite amount of numbers with. Hardest of all is the sheer amount of time it will take to carry out all of these calculations. Unless you happen to have a monkey who is good at these infinite tasks (Google: the Infinite Monkey Theorem), this is quite physically impossible, so we will have to go about it another way.

Let us find a way to visualise the odd numbers - maybe that will make it much clearer of how we should prove this statement.



Above is the first 5 numbers illustrated with blocks for each number. And there's quite an interesting pattern that you might notice. The even numbers (here illustrated with 2 & 4) form squares and rectangles, or, in particular, shapes with lines of symmetry. The odd numbers, however have this one block at the top, which sits alone and awkward. So I can generalise the numbers like this.

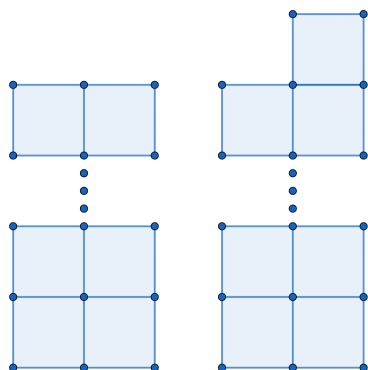
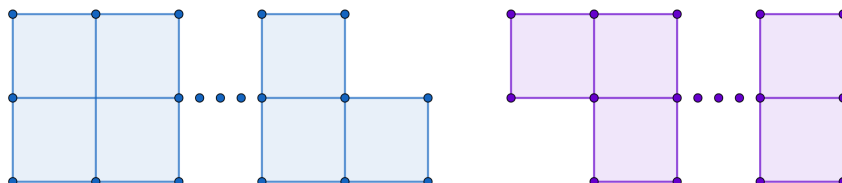


Figure 1: A generic even & odd number.

Now we have an easy way to visualise the difference between an even and an odd number. Quite clearly, an odd number has one extra block. An important note is that those three blue dots are doing a lot of heavy lifting in this diagram. They are acting as a placeholder for all the blocks that have not been drawn physically.

This is crucial, as my even number could be representing 10 or 28, in the same way that my odd number could be representing 7 or 39.

And finally, we now have enough technology and visuals to finalise this proof. We can now see that if we add two different odd numbers together, the lonely block on the top can fit together with the other single block, and like some form of mathematical jigsaw, the addition of these two numbers (or shapes) forms a perfectly symmetrical and tidy rectangle, as demonstrated below.



Finally we can finish the proof off quite happily with either the letters *Q.E.D.* or an empty square, $\square$.

Let us end on a clear example, of the key differences between judicial & scientific proof and our mathematical proof. What better way than to investigate a conjecture (theorem that hasn't been proved yet) from one of the greatest mathematicians of the 18th Century.

Conjecture: (Euler, 1772) The polynomial $n^2 - n + 41$ can be used to generate and find prime numbers.

Now, let us break down this conjecture and play with it somewhat. If there truly is a link between the prime numbers, that would be quite marvellous given their sheer importance in modern day society (Google: RSA Encryption).

In a court of law, if there is a person, who was spotted leaving the premises around the time of the crime, he may be wanted for questioning, but probably won't be scrutinised too heavily. However, if he also happened to be carrying a hammer that fits the description of the murder weapon and had had terse conversation with the victim earlier that day, his lawyers would be worried. The point I am trying to make is that you only need about 4 or 5 pieces of strong circumstantial evidence to be convicted or thoroughly questioned in a criminal case.

Hence, we ought to check the first 5 numbers to test if they are indeed prime.

n	<i>Formula</i>	Prime?
1	$1^2 - 1 + 41 = 41$	Yes
2	$2^2 - 2 + 41 = 43$	Yes
3	$3^2 - 3 + 41 = 47$	Yes
4	$4^2 - 4 + 41 = 53$	Yes
5	$5^2 - 5 + 41 = 61$	Yes

Brilliant! We have 5 strong pieces of evidence, and if I were a lawyer, and not a person of scientific and mathematical standing, I would be completely content with these results. However, this is only a mere 5 values out of an incredibly large amount. The scientist would want to delve further and test more values. And if the scientist tested more values, they would spot a pattern.

Every single number they put in, returns a prime number. Trying $n = 25$ gives 641 which is indeed prime. What about $n = 39$ or $n = 40$? Both work giving 1523 & 1601 which are both prime. At this point, if I were a scientist, I would be completely convinced. There is a clear pattern running through the data and there are 40 pieces of evidence for which this function gives prime numbers

However, as a mathematician, I am still not convinced. It must work for all my possible entries. Hence it must work for an infinite amount of numbers, and if it fails, even for just one number, then the conjecture is disproved. Let us try $n=41$:

$$(41)^2 - (41) + 41 = (41)^2$$

And we have a contradiction!

41^2 has three factors, 1, 41 & 41^2 thus meaning it is not a prime number. Therefore, the proposed conjecture is false. But the key part, is how different the three styles of proof were.

Mathematical proof is the most comprehensive, but also the most interesting and most certain. I hope that this booklet has given you an interesting starter to the world of truth.